



Investigation report

Grounding on Boolarong Bank, Weerona Channel, Kanooka Heads -
Executive brief

Drafted 04/10/2026

INVESTIGATION WORKFLOW SUITE

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What happened

On the evening of 7 July 2026, the passenger ferry MV Tarrakin Star, a 24.4 m aluminium catamaran, was running service TF18 from Muntari Island to Kanooka Heads. It carried 38 passengers and three crew: a master, a deckhand and a kiosk attendant. This was the last service of the day. Under the winter timetable, this return trip had been made in darkness every day since 1 June 2026. The ferry departed at 17:35, after sunset. Earlier that day, at a change of master at about 13:05, the outgoing master told the relieving master that the chart plotter (the electronic chart display used to track the vessel's position) had frozen twice that morning and recovered both times. The fault was not entered in the defect book. The ferry entered the Weerona Channel at 18:03, on the charted centreline at about 8 knots. At 18:04 the deckhand left the bridge to collect fares. At 18:07 the chart plotter froze again and did not recover. The ferry continued at about 8 knots. From about 18:09 its heading was held a few degrees west of the channel's line, and at about that time it crossed the western edge of the dredged channel. From 18:08 the master searched for the No. 4 beacon, a navigation marker at the channel, and sighted it at 18:10. He opened the wheelhouse window during the search in case the glass was the problem. A later check found that the beacon cannot be seen from the channel over a stretch north of it that included where the search began.

At 18:11:48 the echo sounder (the instrument that measures water depth under the hull) sounded its shallow-water alarm. Four seconds later both engines were put astern together. At 18:12:04, 16 seconds after the alarm, the ferry ran onto Boolarong Bank at 6.8 knots, 74 m west of the channel centreline and 54 m outside the dredged channel. It stopped within 11 m, leaning 4 degrees to starboard on a falling tide. The engines were stopped at 18:12:31 and no attempt was made to refloat. At 18:13 the master made an urgency radio call. Port Control acknowledged it and sent the port workboat. All passengers were in lifejackets by about 18:22. Donning was delayed in the forward saloon, where the lifejacket locker was behind the kiosk counter. The workboat came alongside at 18:41, and the last passenger was ashore at 19:26. There were no serious injuries. One passenger reported a grazed shin and two were treated at the wharf for distress. The starboard hull was split and one internal compartment partly flooded. The starboard propeller, shaft and rudder were damaged and the starboard skeg (a fin under the hull) was torn away. The ferry refloated at 00:20 and was withdrawn from service. The channel was closed to commercial traffic for 19 hours.

Why it happened

The ferry grounded because it left the dredged channel in darkness without this being recognised, and the shallowing water under the hull was detected only 16 seconds before contact.

The most significant factor was what happened after the chart plotter froze at 18:07 and did not recover. It was virtually certain that the plotter froze, and extremely likely that this influenced what followed. The ferry continued at about eight knots rather than stopping in the channel. It was extremely likely that continuing at that speed contributed both to the ferry leaving the dredged channel and to the shoaling not being detected until 16 seconds before contact. It was virtually certain that the ferry left the channel and tracked onto Boolarong Bank, and virtually certain that this contributed to the late detection. It was extremely likely that the late detection contributed to the grounding.

The decision to continue was shaped by the operator's arrangements. There was no night passage plan for the Weerona Channel. There was also no hold point (a pre-agreed point at which the vessel stops and waits) for when the primary means of fixing the vessel's position was lost. It was very likely that this contributed to the transit continuing at eight knots (Safety issue). Behind it lay a further gap. When the winter timetable moved the last service into darkness, the route risk assessment was not reopened, and it was likely that this contributed to the absence of a night passage plan and hold point (Safety issue).

The plotter fault had been passed on verbally at the change of master and was not entered in the defect book. It was likely that this contributed to the plotter freezing during the channel transit. The change-of-master defect brief was verbal, with nothing checking that what was said was written down, and it was likely that this contributed to the fault going unrecorded (Safety issue).

With the plotter lost, the master looked for the No. 4 beacon. It was extremely likely that the beacon could not be seen from the channel over the half mile before the vessel left it. It was very likely that this contributed to the vessel leaving the channel. Two safety issues lay behind this. The beacon had been converted to solar power, halving its nominal range (the distance at which it is designed to be seen), and no operator was told. It was likely that this contributed (Safety issue). Clearing of the beacon's sightline had also been deferred, unfunded, for three works rounds, and it was very likely that this contributed (Safety issue). Both stemmed from a wider condition. Aids to navigation on this waterway carry no availability standard and no duty to tell users anything, and it was likely that this contributed to both (Safety issue).

The investigation also examined whether new foreshore floodlighting behind the beacon reduced the master's ability to pick it up. This matter could not be resolved and is not a finding. The floodlighting is aligned behind the beacon, and the masters' accounts were consistent and independent. However, the available evidence could not establish whether the floodlighting affected detection, and the investigation assessed its effect as about as likely as not. The question may be unanswerable while the mangroves remain. Regardless of this outcome, a lighting assessment of the channel is warranted.

What we found

A finding stated without a likelihood term was established as fact.

Contributing safety factors

- When the winter timetable moved the last service into darkness, the route risk assessment was not reopened, and it is likely that this contributed to the occurrence. (Safety issue)
- It is virtually certain that there was no night passage plan for the channel and no hold point (a pre-agreed point at which the vessel stops and waits) if the primary means of fixing the vessel's position was lost, and it is very likely that this contributed. (Safety issue)
- The change-of-master defect brief was verbal, with nothing checking that what was said got written down, and this probably contributed to the occurrence. (Safety issue)
- On the balance of probabilities, the plotter fault being passed on verbally at the change of master and not entered in the defect book contributed to the occurrence.
- With virtual certainty, the chart plotter display froze during the channel transit and did not recover, and this almost certainly contributed.
- The transit continued at eight knots after the plotter froze, rather than stopping in the channel, and it is highly likely that this contributed.
- Aids to navigation on this waterway carry no availability standard and no duty to tell users anything, a condition the evidence establishes conclusively, and it is probable that this contributed. (Safety issue)
- The No. 4 beacon was converted to solar power, its nominal range (the distance at which it is designed to be seen) was halved and no operator was told, and in all likelihood this contributed. (Safety issue)
- Clearing of the beacon's sightline was deferred, unfunded, for three works rounds, and this very probably contributed. (Safety issue)
- It is very likely that the No. 4 beacon was not visible from the channel over the half mile before the vessel left it, and very likely that this contributed.
- The vessel left the dredged channel and tracked onto Boolarong Bank, and this contributed to the grounding.
- It is virtually certain that shoaling (shallowing water under the hull) was not detected until sixteen seconds before contact, and very likely that this contributed.

Other factors that increased risk

- The forward saloon lifejacket locker was not reachable from most of the saloon it serves, and this very likely increased risk. (Safety issue)
- On the day of the occurrence the master was on day nine of twelve, on a split duty across a thirteen hour forty minute spread, and this probably increased risk.

Other findings

- The investigation examined whether a water level below prediction reduced the depth over Boolarong Bank, and ruled it out.
- A steering or propulsion failure causing the vessel to sheer off track was examined and ruled out.

- Stopping the engines and making no attempt to refloat on a falling tide was examined and found to have reduced risk.

What is being done

Aids to navigation on this waterway carry no availability standard and no duty to tell users anything. No action has been recorded against this issue. [NOT ESTABLISHED: whether the body responsible for WAN-2 has been informed, or has proposed any change to section 4.]

The No. 4 beacon was converted to solar power, its nominal range was halved, and no operator was told. The investigation raised its findings on the beacon with the Port Authority and the operator in July 2026, before this report was drafted. [NOT ESTABLISHED: any action taken or proposed in response, including a notice to mariners or correction of the 6 M figure in the operator's passage plan.]

Clearing of the beacon's sightline was deferred, unfunded, for three works rounds. This was raised with the Port Authority at the same time. [NOT ESTABLISHED: whether WO-176 has been funded or the mangroves on Weerona Islet trimmed.]

The winter timetable moved the last service into darkness without the route risk assessment being reopened. On 8 July 2026 the General Manager Operations suspended the evening Muntari services. [NOT ESTABLISHED: whether RRA-07 has since been reviewed, and whether the suspension remains in place.]

There was no night passage plan for the channel and no hold point if the primary means of fixing the vessel's position was lost. Apart from the suspension of evening services, no action has been recorded. [NOT ESTABLISHED: whether a night passage plan or hold point has been written.]

The change-of-master defect brief is verbal, and nothing checks that what is said is written down. No action has been recorded against OPS-9. [NOT ESTABLISHED: whether the plotter firmware issued in 11/2024 has been applied, or whether the operator has subscribed to the manufacturer's bulletins.]

The forward saloon lifejacket locker cannot be reached from most of the saloon it serves. The same counter is fitted to the operator's two other 24 m vessels. The question of the whole fleet is with the operator and the state regulator, and their answers are awaited.

Distribution

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